Problem Statement

Excel Insights Engine

Background

Organizations often receive large and complex Excel files containing multiple sheets of structured or semi-structured data. Manually exploring these files to understand column types, identify missing values, detect outliers, analyze categorical and date fields, and generate meaningful summaries can be time-consuming and error-prone.

Objective

The objective is to develop an **automated Excel Insights Engine** that can:

1	Load Excel files with multiple sheets.
2	Provide essential metadata such as row and column counts, column names, and data types.
3	Detect and quantify missing values.
4	Generate summary statistics for numeric columns.
5	Identify strong correlations between numeric variables.
6	Detect outliers using statistical methods like IQR or Z-score.
7	Analyze categorical columns for frequency distributions and data quality.
8	Identify and summarize patterns in date columns, including day, month, and year distributions.
9	Automatically create visualizations to help users interpret the data.
10	Compile all findings into a structured report in markdown format or plain text.

Who it is for

This tool will be useful for **data analysts, auditors, business users, and domain experts** who need to extract quick insights from Excel data without writing manual code every time.